

Quantum Bridge Symmetric Key Distribution System

Unbreakable key distribution technology for protecting data and infrastructure

Overview

The Quantum Bridge Symmetric Key Distribution System (SDS) provides quantum-safe symmetric keys via SKIP and ETSI API (GS QKD 014). The SDS provides a flexible and comprehensive security solution that enables organizations to protect their sensitive data and communications across the entire network stack, regardless of the encryption methods or devices used.

The number of SDS nodes in the network can be scaled dynamically without impacting the performance, security, or usability of the entire solution, making it an ideal choice for organizations with growing or varying security needs. This scalability ensures that the platform can adapt to changing requirements without compromising its core functions or user experience.

Our solution delivers quantum-safe, crypto-agile security for critical data and communications. Supporting Post-Quantum Cryptography (PQC), Quantum Key Distribution (QKD), Pre-Shared Keys (PSK), and the Distributed Symmetric Key Establishment (DSKE) protocol for unconditional security, the SDS protects against even the most advanced threats, ensuring that organizations can meet their security and compliance requirements with confidence.

Key Management Entity

The Quantum Bridge Key Management Entity (KME) is the core element of the SDS. Each KME can deliver symmetric keys to multiple network appliances for use at any layer, for example, network encryptors, firewalls and routers.

Key Features

- Layer-agnostic
- Compatible with existing networks via ETSI GS QKD 014 or SKIP
- Crypto-agility (PQC, PSK, QKD, DSKE)
- Distributed trust architecture, with no single point of failure
- Perfect secrecy and quantum-safety
- Automated dynamic symmetric key distribution
- Up to 100 appliances per KME
- Multi-layer security
- Supports SNMP or Syslog protocol for monitoring

Applications

- **Layer 1:** Optical encryptors
- **Layer 2:** MACsec encryptors, routers, switches
- **Layer 3:** IPsec encryptors, WireGuard®, VPNs, compatible with RFC 8784

Cryptography

- **Post-Quantum Cryptography (PQC):**
 - CRYSTALS-Kyber (512, 768, and 1024)
 - ML-KEM (512, 768, and 1024)
 - BIKE (Level 5)
 - Classic McEliece (mceliece6960119)
- **DSKE:**
 - Information-theoretic security
 - Confidentiality: One-time-pad encryption between KME and Security Hub
 - Authentication: One-time key with universal hash function
 - Trust distribution: Shamir secret sharing
 - AES-256-GCM for anti-DoS message security
- **Quantum Key Distribution (QKD):**
 - Supported for additional link security

Use Cases

WAN/LAN interconnect. The Quantum Bridge SDS provides cryptographic keys to protect all valuable information. The Quantum Bridge KME can be interfaced with most existing network appliances to build secure communication networks. The solution can be used to protect traffic between different WAN/LAN networks, especially when communication goes through untrusted third-party infrastructure or the public internet.



Data centre interconnect. The Quantum Bridge SDS provides *information-theoretic security*, which implies security even against adversaries with unlimited quantum or classical computational resources. The Quantum Bridge SDS is layer-agnostic and can integrate into Layers 1, 2, and 3, interfacing with existing network appliances, or running as a native application in existing hardware. The SDS can be used to provide secure access to data centres and ensure the highest level of security for both encryption and authentication.

Custom Development

The Quantum Bridge SDS software can be tailored to meet users' specific needs or requirements. Some use cases include:

- **Radio communication**
- **Satellite communication**
- **Sensor communication**

Please contact Quantum Bridge to discuss use cases of interest or custom development to integrate the Quantum Bridge SDS into your infrastructure.

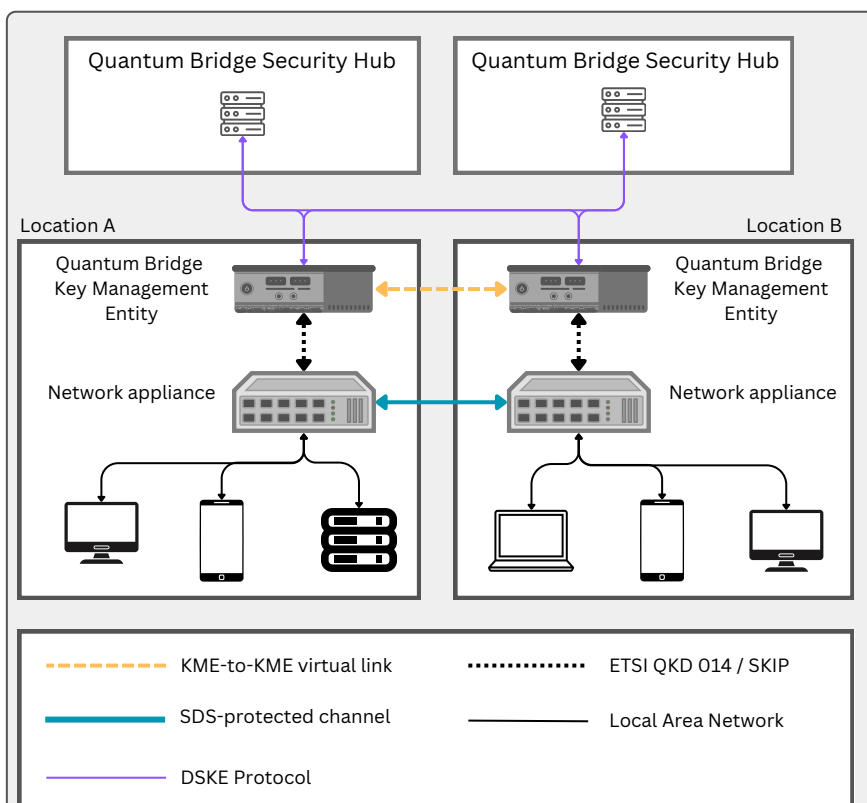


Implementation

The Quantum Bridge SDS software is developed in Rust, a programming language distinguished for its small footprint, enhanced security, and unparalleled performance. Rust's emphasis on memory- and thread-safety makes it ideal for creating secure applications, effectively minimizing vulnerabilities and ensuring robustness. This ensures that our software uses minimal resources while maximizing execution speed, setting it apart in efficiency.

Deployment

Cryptographic keys supplied by the KME to existing appliances can be combined with other keys in compatible existing protocols to make current infrastructure quantum-safe and future-proof, and to eliminate any vulnerability associated with asymmetric key management protocols or Public Key Infrastructure (PKI) and its certificates.



Example deployment. In the figure, Quantum Bridge KMEs in different locations are connected to network appliances to provide quantum-safe keys. A network appliance, such as a firewall, network encryptor or router, must support ETSI GS QKD 014 or SKIP. The diagram also includes two Security Hubs, needed should the DSKE protocol be activated on top of PQC. PQC and DSKE independently provide end-to-end security of key delivery. Security of traffic between KMEs is additionally protected through a Virtual Private Network (VPN), where the link security can be further enhanced by PSK or QKD. All traffic is delivered over the VPN, where packets are routed from KME to KME with flexible routing rules.

Software Requirements

- **Host**
 - Operating System*: RHEL 9.4; Ubuntu 22.04
- **CPU**
 - Minimum: Intel® Core™ i3 (2 GHz) or equivalent
 - Recommended: Intel® Core™ i5 processor or equivalent
- **Memory**
 - Minimum: 4 GiB RAM
 - Recommended: ≥ 8 GiB RAM
- **Disk**
 - Minimum 5 GB
 - Additional storage for DSKE
- **Network**
 - UDP + TCP + link management (DNS, etc.)
 - IP connectivity for the following links:
 - KME to KME
 - KME to Security Hub (if DSKE is activated)

Performance

- **KME key throughput (256-bit keys)**
 - > 12,000 keys per second in PQC mode
 - > 1,000 keys per second in DSKE mode
- **KME average response time to network appliance**
 - Max load: 50 ms
 - Low load: 20 ms
- **Security Hub key request response time**
 - < 100 ms (excluding network delays)

Routing Capabilities

Configuration of the KME virtual private network allows for the definition of:

- Static routes between specific KME appliances
- Dynamic routing using OSPF (default), IS-IS, RIP or any protocol supported by the FRRouting Project
- Routing cost metrics based on the security paradigm of each tunnel (QKD or PSK)
- SNMP hosts to send traps to as well as the traps to send

* Available on Debian-based Linux distributions on request.